Individual Assignment

Activeness-Based Clustering of Bank of Twizel Clients

BUSINFO – 700 Analysis of Business Problems

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Executive Summary

This segmentation initiative analyses 9,800 customers of the Bank of Twizel to identify patterns in product usage and customer activeness, aiming to enable differentiated engagement strategies. A Two-Step clustering approach (log-likelihood distance) was applied using six activeness-related variables, resulting in four distinct and statistically valid customer segments.

- Passive & Less Active Clients (46.5%) with minimal transactions, low product ownership, and low balances.

  Launch reactivation campaigns and basic digital onboarding. Offer bundled starter products (e.g., no-fee credit cards, savings boosters). Implement inactivity tracking and low-cost engagement incentives.

- Active Credit Clients (17.7%) engage in high credit card activity, moderate balances, and frequent transactions.

  Expand loyalty and cashback rewards programs and offer personalized credit expansion options. Promote app-based, mobile-first banking experiences.

- High-Valued Premium Clients (17.5%) maintain high balances, diversify product usage, and are financially engaged.

  Assign dedicated banking advisors and VIP service tiers so we upsell investment and wealth management offerings. Focus retention via priority access, exclusive events, and tailored communication.

- Loan-Engaged Actives (18.2%) have high loan ownership, moderate balances, and low transactions. Also, they hold about 2–3 products.

  Cross-sell secured cards and top-up loans. Bundle EMI-linked accounts or salary-linked benefits. Also, protection plans and flexible repayment features should be introduced.

This segmentation equips leadership with a targeted roadmap for strategic marketing, cross-selling, retention, and profitability improvement, focused on optimizing value from the existing customer base.

Overview

This assignment focuses on segmenting the Bank of Twizel's customers using a Two-Step clustering method. The objective is to identify patterns in customer activity and product usage, allowing the bank to group customers into segments. These insights develop targeted marketing strategies to improve customer engagement, retention, and overall profitability.

Data Understanding and Preparation

We began with descriptive statistics to ensure data quality across the 9,800 customer records, which included both categorical and continuous variables.

- Categorical variables (e.g., owns a loan, receives student allowance) were analysed using frequency and percentage.
- Continuous variables (e.g., age, total products, Number of transactions) were assessed using mean, median, minimum, and maximum.

Technique Used *Univariate Descriptive Analysis*

- Results showed no missing values, no major outliers, and no skewed data, confirming the dataset was clean.
- This analysis served as a clustering baseline, helping us with variable distributions.

The data was grouped into three sets: Product ownership (binary), Balance and transaction data (continuous), and Nominal attributes (e.g., loan, auto savings) (Refer to Appendix A).

Variable Selection and Justification

To build meaningful customer clusters, we selected variables that reflect customer activeness and product engagement. The process included both initial testing and refinement based on data relevance and statistical significance.

1. Initial Analysis of listed variables

We analysed variables, including the Number of transactions, Total products owned, Automatic savings, Owns loan, Receiving student allowance, and Balances in checking and savings accounts.

Clusters with less than 7% of the data were removed due to insufficient size for reliable analysis (see Appendix B(i)).

2. Variables Excluded from Clustering

These were removed due to low relevance or data quality issues

- Age and other product ownership binaries – no strong link to activeness; risk of skewing results
- Balance variables with mostly zeros – low variation, limited values
- Student allowance –applies to younger customers but not to other age groups.
- Automatic savings – removed to reduce complexity

3. Final Variables considered in Clustering

- Number of transactions – a direct measure of activity
- Total products owned – reflects overall engagement
- Owns a loan – indicates credit involvement
- Owns a credit card – linked to spending behaviour
- Total Balance and checking account balance – show financial value and usage.

These variables were chosen for their relationship to activeness (see Appendix B(ii)).

4. Statistical Validation

Chi-square tests were applied to key categorical variables (e.g., owns a loan, total products) to confirm they were not redundant and had strong relationships with clustering outcomes (see Appendix C). This helped ensure **model accuracy and robustness**.

Clustering Methodology and Execution

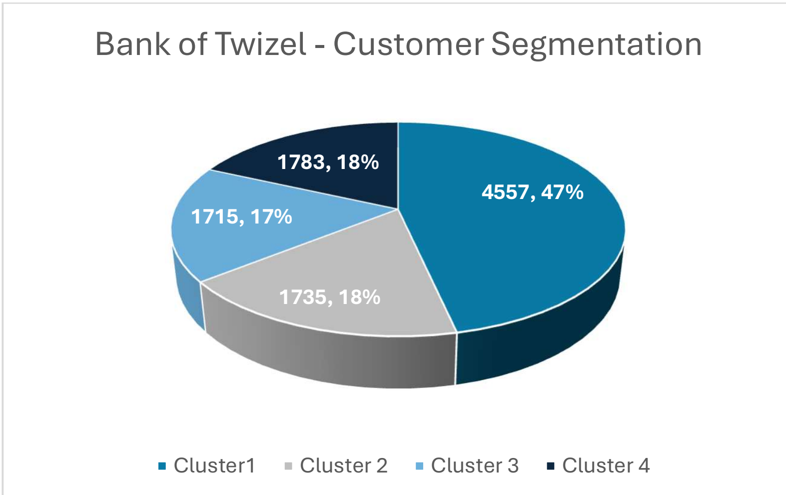
We used Two-Step cluster analysis with log-likelihood distance because it handles both categorical & continuous variables, which suited our mixed dataset of 9,800 customers.

Unlike K-Means (which only works with continuous data using Euclidean distance), Two-Step can accurately process variables like "Owns Loan" (categorical) and "Number of Transactions" or "Total Balance" (continuous). Using Euclidean distance in this case would have distorted the results.

The model used six input variables and produced four meaningful clusters, each of which was of good quality (cluster quality > 0.5).

Cluster Size and Distribution Overview

The Two-Step analysis worked well for our mixed dataset, producing balanced clusters that were not too small (<7%) or overly dominant (>50%).



- Cluster segment 1 is the largest group and includes passive, low-engagement customers.
- Each cluster reflects unique patterns of transactions and product usage.
- After clustering, we generated a cluster membership variable in SPSS to label each customer from Cluster 1 to Cluster 4, enabling further analysis and profiling.

Table 1 Cluster Summary Table (*Refer to Appendix E for the clusters and cluster comparisons*)

Cluster	**Size**	**Key Traits**
Cluster 1	46.5%	Low balance, few products, low transactions
Cluster 2	17.7%	High transactions, Active credit card users
Cluster 3	17.5%	High balances, Multiple products, financially engaged
Cluster 4	18.2%	Own loans, medium product use, moderate transactions

Cluster Profiling and Evaluation of Cluster Significance

To ensure the quality and relevance of our identified clusters, we assessed the significance of both categorical and continuous variables.

1. Categorical Variables – Chi-Square Tests

- Strong associations were found between clusters and key variables like "Total Products Owned" and "Owns a Loan" ($p < 0.05$), confirming their importance in defining customer activeness.
- Variables like "Receives Student Allowance" and "Has Automatic Savings" showed no strong link to activeness ($p > 0.05$) and were rightly excluded.
- This helped distinguish active variables (impactful for clustering) from passive ones (not useful for segmentation).

2. Continuous Variables – ANOVA and Post Hoc Testing

- ANOVA confirmed that the Number of Transactions, Total Balance, and Checking Balance varied significantly across clusters ($p < 0.05$).
- Levene's Test showed unequal variance, justifying Tamhane's T2 post hoc test.
- Post hoc results confirmed that clusters are statistically distinct on these variables.
- Passive variables like age, share balances, and certificates were not significant, supporting their exclusion.

These tests validate that the chosen variables effectively differentiate customer segments, supporting the robustness and reliability of the clustering model.

Variable	Test Applied	p-value	Significance/ Relevance	Inclusion in Clustering
Products Owned	Chi-Square	< 0.05	Significant	Included (Active)
Owns a Loan	Chi-Square	< 0.05	Significant	Included (Active)
Student Allowance	Chi-Square	< 0.05	Not relevant	Excluded (Passive)
Automatic Savings	Chi-Square	< 0.05	Not relevant	Excluded (Passive)
Number of Transactions	ANOVA + Tamhane	< 0.05	Significant	Included (Active)
Total Account Balance	ANOVA + Tamhane	< 0.05	Significant	Included (Active)
Checking Account Balance	ANOVA + Tamhane	< 0.05	Significant	Included (Active)

Cluster Interpretation and Labelling based on SPSS Output on Clusters

We have identified the following customer segments based on their activity and product usage patterns

Cluster 1 – Passive Clients (46.5%)

- Very low transactions and balances. A few products (1–2) rarely use bank services.
- Little to no credit card or loan ownership

Insight: Largest group but least active. Low contribution and minimal engagement.

Cluster 2 – Active Credit Clients (17.7%)

- High transaction volume, Frequent credit card usage
- Moderate account balance, owning 2–3 products.

Insight: Credit-focused, digitally active customers. Great fit for rewards and credit promotions.

Cluster 3 – Premium Clients (17.5%)

- High account balance, owning 3+ products, Own loans, and credit cards
- Moderate to high transaction activity

Insight: Financially valuable and well-engaged. Key group for retention and upselling.

Cluster 4 – Loan-Engaged Clients (18.2%)

- High loan ownership, Moderate transactions, and balances.
- 2–3 product ownership and credit card usage patterns.

Insight: Loan-based relationship. Suitable for bundled services and financial support tools.

Each cluster shows unique engagement patterns. Tailored strategies will help increase product usage, strengthen relationships, and grow customer value.

Strategic Recommendations by Customer Segment

For the above cluster segmentations, we would like to propose strategies and proposals to maximize profit potential based on customer behaviour and patterns.

Cluster 1 Passive Clients (46.5%)

Account reactivation nudges, bundled starter products, and inactivity incentives will engage low-active customers.

Low engagement requires low-barrier re-entry points. Even small reactivation boosts can yield large returns due to the size of this group.

Cluster 2 Active Credit Clients (17.7%)

Enhanced card rewards points, pre-approved credit offers, and the introduction of a mobile app are measures to increase cash inflows and expenditures.

Credit-heavy behaviour aligns with loyalty and credit-based upselling. Mobile engagement supports digital-savvy habits and re-engages customers with more online product offers.

Cluster 3 Premium Clients (17.5%)

Personal advisors, an advanced investment scheme, and exclusive perks offer a premium feel.

High account balances and product use indicate wealth potential targeted retention and upsell of the bank's profit-generating products.

Cluster 4 Loan-Engaged Clients (18.2%)

Cross-sell secured credit, EMI-linked accounts, and financial planning tools will aid this segment.

Loan focus makes them receptive to structured financial products and supportive financial services.

Appendix – References

Appendix – A – Descriptive Statistics – Frequency and missing values

Frequencies

Statistics

		Owns checkings account	Owns Shares	Owns investment trust	Owns certificate	Owns money deposit	Owns savings account	Owns mortgage	Owns Credit card	Has income at this bank
N	Valid	9800	9800	9800	9800	9800	9800	9800	9800	9800
	Missing	0	0	0	0	0	0	0	0	0

Frequency Table

Owns checkings account

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	.00	3689	37.6	37.6	37.6
	1.00	6111	62.4	62.4	100.0
	Total	9800	100.0	100.0	

Owns Shares

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	.00	9736	99.3	99.3	99.3
	1.00	64	.7	.7	100.0
	Total	9800	100.0	100.0	

Frequencies

Statistics

		Balance on checkings account	Balance on shares	Balance on savings account	Balance on investment trust	Balance on certificate	Balance on money deposit	Balance on mortgage	Balance on loan	Age in years	Number of transactions	tot_balance
N	Valid	9800	9800	9800	9800	9800	9800	9800	9800	9800	9800	9800
	Missing	0	0	0	0	0	0	0	0	0	0	0
Mean		1743.5609	159.6734	3576.2087	652.6573	421.2353	5.1020	3625.9636	641.0561	42.9905	26.4213	10948.5557
Median		15.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	41.0000	1.0000	589.5000
Mode		.00	.00	.00	.00	.00	.00	.00	.00	39.00	.00	.00
Minimum		-14193.00	.00	-184.00	.00	.00	.00	.00	.00	.00	.00	.00
Maximum		139465.00	287096.00	525490.00	236143.00	250000.00	25000.00	566930.00	100000.00	113.00	282.00	566930.00

Frequencies

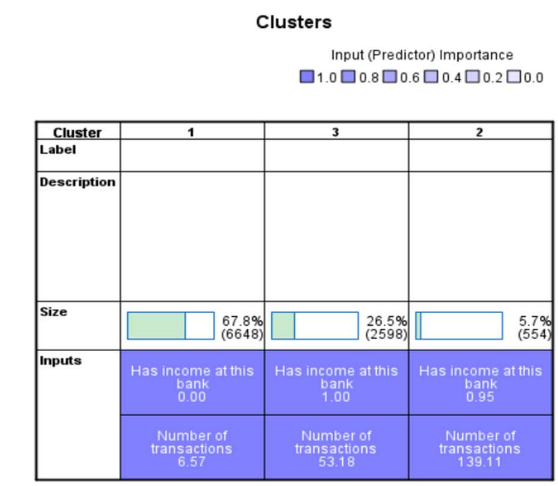
Statistics

		Owns damage insurance	Owns loan	Recieves student allowance	Automatic saving	tot_prod	response
N	Valid	9800	9800	9800	9800	9800	9800
	Missing	0	0	0	0	0	0

Appendix – B (i)

The cluster below is identified during the initial analysis, where cluster quality is moderately low, and size is less than 7%

TwoStep Cluster



Clusters

Input (Predictor) Importance

1.0 0.8 0.6 0.4 0.2 0.0

Cluster	1	3	2
Label			
Description			
Size	67.8% (6648)	26.5% (2598)	5.7% (554)
Inputs	Has income at this bank 0.00	Has income at this bank 1.00	Has income at this bank 0.95
	Number of transactions 6.57	Number of transactions 53.18	Number of transactions 139.11

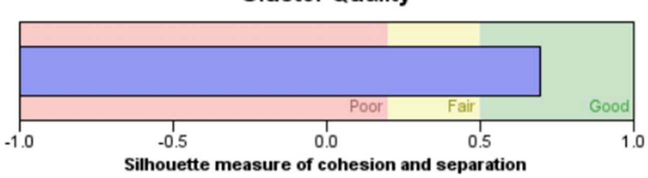
Appendix – B (ii) – Quality Cluster by refining variables

TwoStep Cluster

Model Summary

Algorithm	TwoStep
Inputs	6
Clusters	4

Cluster Quality



Input (Predictor) Importance

1.0 0.8 0.6 0.4 0.2 0.0

Cluster	1	2	3	4
Label				
Description				
Size	46.5% (4561)	17.7% (1736)	17.5% (1717)	18.2% (1786)
Inputs	Number of transactions 13.29	Number of transactions 70.40	Number of transactions 0.06	Number of transactions 42.55
	Owns Credit card 0.00	Owns Credit card 0.47	Owns Credit card 0.00	Owns Credit card 0.00
	Owns loan 0 (100.0%)	Owns loan 0 (77.0%)	Owns loan 0 (100.0%)	Owns loan 0 (100.0%)
	tot_balance 1,775.44	tot_balance 46,599.48	tot_balance 0.00	tot_balance 10,247.14
	tot_prod 1 (100.0%)	tot_prod a least 3 (73.8%)	tot_prod Nothing (100.0%)	tot_prod 2 (100.0%)
	Balance on checkings account 786.73	Balance on checkings account 5,590.57	Balance on checkings account 0.00	Balance on checkings account 2,123.96

Appendix – C

Crosstabs

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Owns loan * tot_prod	9800	100.0%	0	0.0%	9800	100.0%

Owns loan * tot_prod Crosstabulation

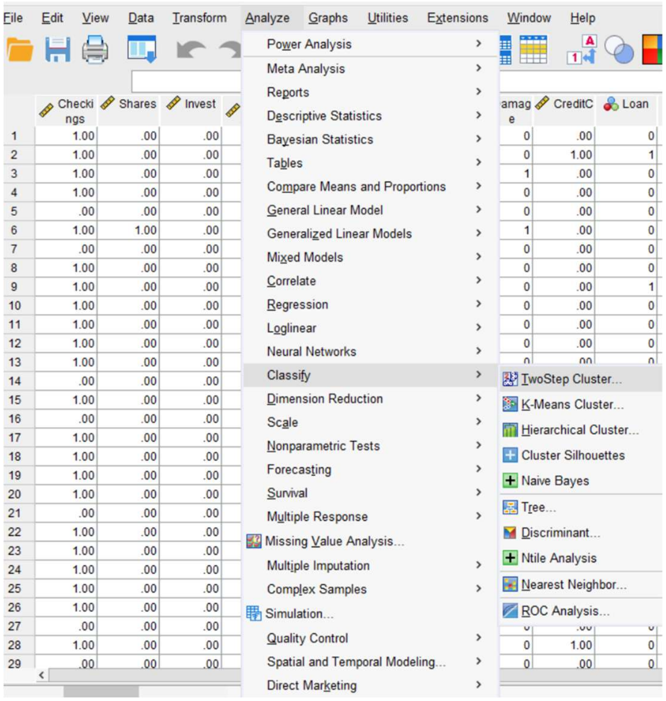
			tot_prod				
			Nothing	1	2	a least 3	Total
Owns loan	0	Count	1717	4663	2054	966	9400
		% within Owns loan	18.3%	49.6%	21.9%	10.3%	100.0%
	1	Count	0	1	83	316	400
		% within Owns loan	0.0%	0.3%	20.8%	79.0%	100.0%
Total		Count	1717	4664	2137	1282	9800
		% within Owns loan	17.5%	47.6%	21.8%	13.1%	100.0%

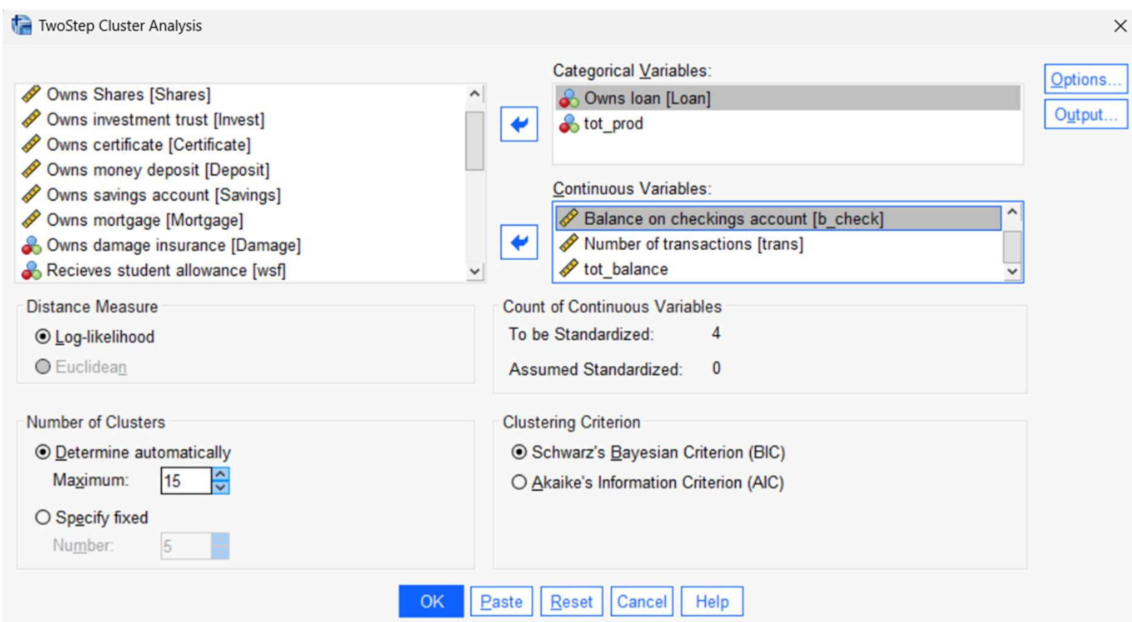
Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	1654.855[a]	3	<.001
Likelihood Ratio	1189.680	3	<.001
Linear-by-Linear Association	1113.196	1	<.001
N of Valid Cases	9800		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 52.33.

Appendix – D


File Edit View Data Transform Analyze Graphs Utilities Extensions Window Help
Power Analysis
Meta Analysis
Reports
Descriptive Statistics
Bayesian Statistics
Tables
Compare Means and Proportions
General Linear Model
Generalized Linear Models
Mixed Models
Correlate
Regression
Loglinear
Neural Networks
Classify
Dimension Reduction
Scale
Nonparametric Tests
Forecasting
Survival
Multiple Response
Missing Value Analysis...
Multiple Imputation
Complex Samples
Simulation...
Quality Control
Spatial and Temporal Modeling...
Direct Marketing
TwoStep Cluster...
K-Means Cluster...
Hierarchical Cluster...
Cluster Silhouettes
Naive Bayes
Tree...
Discriminant...
Ntile Analysis
Nearest Neighbor...
ROC Analysis...

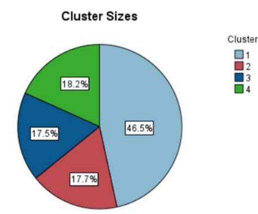

TwoStep Cluster Analysis
Owns Shares [Shares]
Owns investment trust [Invest]
Owns certificate [Certificate]
Owns money deposit [Deposit]
Owns savings account [Savings]
Owns mortgage [Mortgage]
Owns damage insurance [Damage]
Recieves student allowance [wsf]
Categorical Variables:
Owns loan [Loan]
tot_prod
Continuous Variables:
Balance on checkings account [b_check]
Number of transactions [trans]
tot_balance
Options...
Output...
Distance Measure
Log-likelihood
Euclidean
Count of Continuous Variables
To be Standardized: 4
Assumed Standardized: 0
Number of Clusters
Determine automatically
Maximum: 15
Specify fixed
Number: 5
Clustering Criterion
Schwarz's Bayesian Criterion (BIC)
Akaike's Information Criterion (AIC)
OK
Paste
Reset
Cancel
Help

Appendix – E

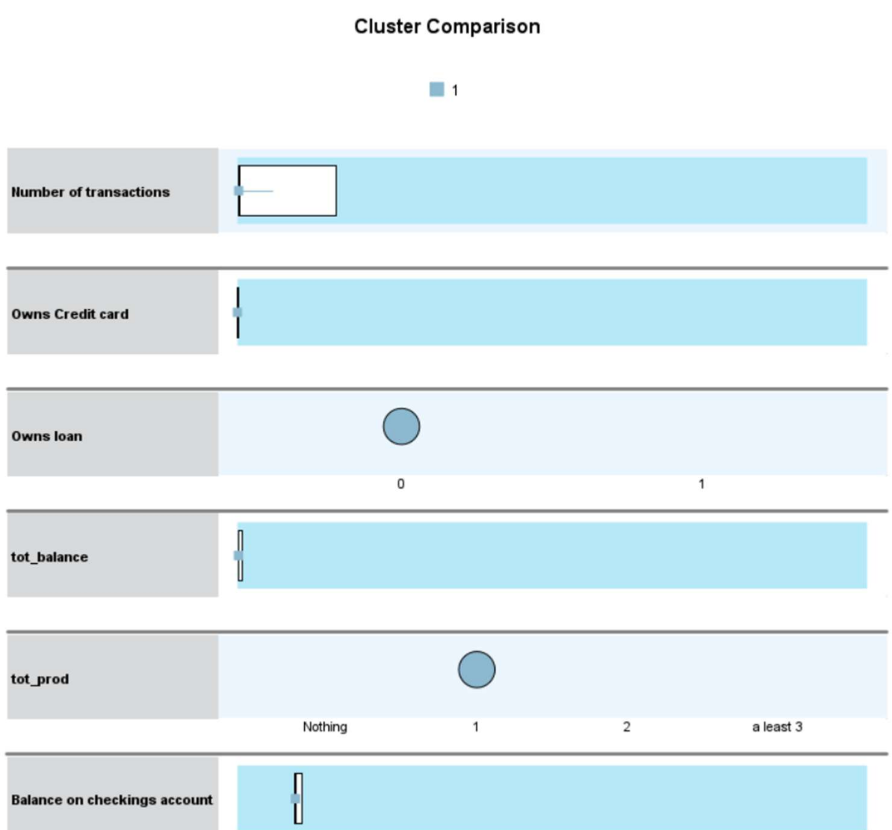
Clusters

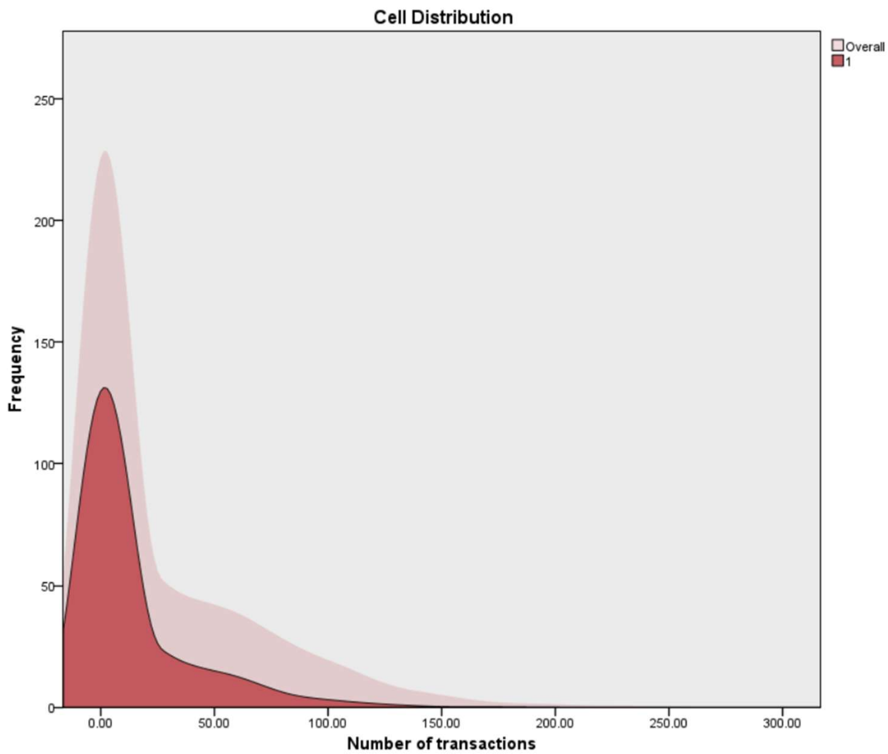
Input (Predictor) Importance
1.0 0.8 0.6 0.4 0.2 0.0

Cluster	1	2	3	4
Label				
Description				
Size	46.5% (4561)	17.7% (1736)	17.5% (1717)	18.2% (1786)
Inputs	Number of transactions	Number of transactions	Number of transactions	Number of transactions
	Owns Credit card 0.00	Owns Credit card 0.47	Owns Credit card 0.00	Owns Credit card 0.00
	Owns loan 0 (100.0%)	Owns loan 0 (77.0%)	Owns loan 0 (100.0%)	Owns loan 0 (100.0%)
	tot_balance 1,775.44	tot_balance 46,599.48	tot_balance 0.00	tot_balance 10,247.14
	tot_prod 1 (100.0%)	tot_prod a least 3 (73.8%)	tot_prod Nothing (100.0%)	tot_prod 2 (100.0%)
	Balance on checkings account	Balance on checkings account	Balance on checkings account	Balance on checkings account

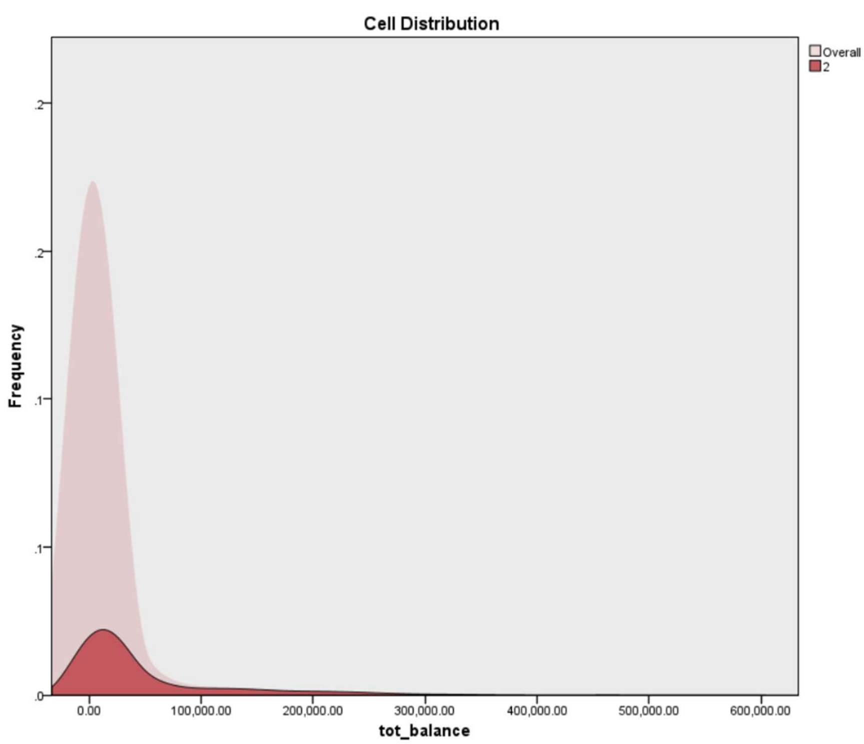


Size of Smallest Cluster	1717 (17.5%)
Size of Largest Cluster	4561 (46.5%)
Ratio of Sizes: Largest Cluster to Smallest Cluster	2.66





Cell Distribution
Overall
1
Frequency
250
200
150
100
50
0
0.00
50.00
100.00
150.00
200.00
250.00
300.00
Number of transactions



Cell Distribution
Overall
2
Frequency
.2
.2
.1
.1
.0
0.00
100,000.00
200,000.00
300,000.00
400,000.00
500,000.00
600,000.00
tot_balance

Appendix – F

Oneway

Descriptives

		N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean Lower Bound	95% Confidence Interval for Mean Upper Bound	Minimum	Maximum
Owns Credit card	1	4561	.0000	.00000	.00000	.0000	.0000	.00	.00
	2	1736	.4724	.49938	.01199	.4488	.4959	.00	1.00
	3	1717	.0000	.00000	.00000	.0000	.0000	.00	.00
	4	1786	.0000	.00000	.00000	.0000	.0000	.00	.00
	Total	9800	.0837	.27691	.00280	.0782	.0892	.00	1.00
Balance on checkings account	1	4561	786.7314	2273.33398	33.66148	720.7386	852.7242	-14193.00	21187.00
	2	1736	5590.5685	11304.77577	271.32330	5058.4134	6122.7237	-6539.00	139465.00
	3	1717	.0000	.00000	.00000	.0000	.0000	.00	.00
	4	1786	2123.9574	2971.51574	70.31327	1986.0525	2261.8624	-3936.00	17493.00
	Total	9800	1743.5609	5499.84302	55.55680	1634.6581	1852.4637	-14193.00	139465.00
tot_balance	1	4561	1775.4357	6402.89948	94.80836	1589.5653	1961.3060	.00	126000.00
	2	1736	46599.4781	72413.18127	1737.97197	43190.7377	50008.2185	.00	566930.00
	3	1717	.0000	.00000	.00000	.0000	.0000	.00	.00
	4	1786	10247.1389	16276.13276	385.13279	9491.7803	11002.4974	1.00	115791.00
	Total	9800	10948.5557	35793.89043	361.57289	10239.7983	11657.3131	.00	566930.00
Number of transactions	1	4561	13.2903	24.91772	.36896	12.5669	14.0136	.00	181.00
	2	1736	70.3957	49.79689	1.19516	68.0516	72.7398	.00	282.00
	3	1717	.0600	.90865	.02193	.0170	.1030	.00	30.00
	4	1786	42.5543	39.24999	.92875	40.7328	44.3759	.00	204.00
	Total	9800	26.4213	40.01016	.40416	25.6291	27.2136	.00	282.00

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
Owns Credit card	Based on Mean	875948.990	3	9796	<.001
	Based on Median	2405.306	3	9796	<.001
	Based on Median and with adjusted df	2405.306	3	1735.000	<.001
	Based on trimmed mean	709035.122	3	9796	<.001
Balance on checkings account	Based on Mean	701.242	3	9796	<.001
	Based on Median	393.703	3	9796	<.001
	Based on Median and with adjusted df	393.703	3	2354.504	<.001
	Based on trimmed mean	483.676	3	9796	<.001
tot_balance	Based on Mean	2004.576	3	9796	<.001
	Based on Median	840.838	3	9796	<.001
	Based on Median and with adjusted df	840.838	3	2001.681	<.001
	Based on trimmed mean	1415.269	3	9796	<.001
Number of transactions	Based on Mean	1338.069	3	9796	<.001
	Based on Median	1000.927	3	9796	<.001
	Based on Median and with adjusted df	1000.927	3	7763.179	<.001
	Based on trimmed mean	1233.501	3	9796	<.001

ANOVA

		Sum of Squares	df	Mean Square	F	Sig.
Owns Credit card	Between Groups	318.715	3	106.238	2405.306	<.001
	Within Groups	432.673	9796	.044		
	Total	751.388	9799			
Balance on checkings account	Between Groups	35345699875	3	11781899958	442.108	<.001
	Within Groups	2.611E+11	9796	26649359.945		
	Total	2.964E+11	9799			
tot_balance	Between Groups	2.797E+12	3	9.323E+11	935.979	<.001
	Within Groups	9.758E+12	9796	996078102.08		
	Total	1.255E+13	9799			
Number of transactions	Between Groups	5801440.502	3	1933813.501	1916.417	<.001
	Within Groups	9884922.841	9796	1009.077		
	Total	15686363.343	9799			

Appendix - G

Post Hoc Tests

Multiple Comparisons

Dependent Variable		(I) TwoStep Cluster Number	(J) TwoStep Cluster Number	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval Lower Bound	Upper Bound
Owns Credit card	Tamhane	1	2	-.47235*	.01199	<.001	-.5039	-.4408
			3	.00000	.00000	.	.0000	.0000
			4	.00000	.00000	.	.0000	.0000
		2	1	.47235*	.01199	<.001	.4408	.5039
			3	.47235*	.01199	<.001	.4408	.5039
			4	.47235*	.01199	<.001	.4408	.5039
		3	1	.00000	.00000	.	.0000	.0000
			2	-.47235*	.01199	<.001	-.5039	-.4408
			4	.00000	.00000	.	.0000	.0000
		4	1	.00000	.00000	.	.0000	.0000
			2	-.47235*	.01199	<.001	-.5039	-.4408
			3	.00000	.00000	.	.0000	.0000
Balance on checkings account	Tamhane	1	2	-4803.83713*	273.40342	<.001	-5523.9693	-4083.7049
			3	786.73142*	33.66148	<.001	698.1283	875.3345
			4	-1337.22603*	77.95545	<.001	-1542.4836	-1131.9684
		2	1	4803.83713*	273.40342	<.001	4083.7049	5523.9693
			3	5590.56855*	271.32330	<.001	4875.8908	6305.2463
			4	3466.61110*	280.28609	<.001	2728.4245	4204.7977
		3	1	-786.73142*	33.66148	<.001	-875.3345	-698.1283
			2	-5590.56855*	271.32330	<.001	-6305.2463	-4875.8908
			4	-2123.95745*	70.31327	<.001	-2309.1598	-1938.7551
		4	1	1337.22603*	77.95545	<.001	1131.9684	1542.4836
			2	-3466.61110*	280.28609	<.001	-4204.7977	-2728.4245
			3	2123.95745*	70.31327	<.001	1938.7551	2309.1598
tot_balance	Tamhane	1	2	-44824.04246*	1740.55600	<.001	-49408.7142	-40239.3707

Insignificance identified on Passive variables.

		3	1	.00000	.00000	.	.0000	.0000
			2	-3451.77880*	381.08246	<.001	-4455.5671	-2447.9905
			4	-226.06607*	72.80328	.012	-417.8270	-34.3051
		4	1	226.06607*	72.80328	.012	34.3051	417.8270
			2	-3225.71273*	387.97443	<.001	-4247.5754	-2203.8500
			3	226.06607*	72.80328	.012	34.3051	417.8270
Balance on certificate	Tamhane	1	2	-1629.65917*	266.21008	<.001	-2330.8493	-928.4690
			3	105.17781*	29.25160	.002	28.1823	182.1733
			4	-251.32443*	81.96177	.013	-467.1528	-35.4960
		2	1	1629.65917*	266.21008	<.001	928.4690	2330.8493
			3	1734.83698*	264.59809	<.001	1037.8737	2431.8003
			4	1378.33474*	275.45277	<.001	652.8981	2103.7714
		3	1	-105.17781*	29.25160	.002	-182.1733	-28.1823
			2	-1734.83698*	264.59809	<.001	-2431.8003	-1037.8737
			4	-356.50224*	76.56419	<.001	-558.1693	-154.8352
		4	1	251.32443*	81.96177	.013	35.4960	467.1528
			2	-1378.33474*	275.45277	<.001	-2103.7714	-652.8981
			3	356.50224*	76.56419	<.001	154.8352	558.1693
Balance on shares	Tamhane	1	2	-845.08509*	243.92579	.003	-1487.5966	-202.5736
			3	.00132	.00132	.899	-.0021	.0048
			4	-54.71481	32.84740	.454	-141.2336	31.8039
		2	1	845.08509*	243.92579	.003	202.5736	1487.5966
			3	845.08641*	243.92579	.003	202.5749	1487.5979
			4	790.37028*	246.12749	.008	142.0853	1438.6552
		3	1	-.00132	.00132	.899	-.0048	.0021
			2	-845.08641*	243.92579	.003	-1487.5979	-202.5749
			4	-54.71613	32.84740	.454	-141.2349	31.8026
		4	1	54.71481	32.84740	.454	-31.8039	141.2336
			2	-790.37028*	246.12749	.008	-1438.6552	-142.0853
			3	54.71613	32.84740	.454	-31.8026	141.2349

*. The mean difference is significant at the 0.05 level.

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	25349.326[a]	9	<.001
Likelihood Ratio	22150.406	9	<.001
Linear-by-Linear Association	242.220	1	<.001
N of Valid Cases	9800		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 224.61.

TwoStep Cluster Number * Owns loan

Crosstab

Count

		Owns loan		
		0	1	Total
TwoStep Cluster Number	1	4561	0	4561
	2	1336	400	1736
	3	1717	0	1717
	4	1786	0	1786
Total		9400	400	9800

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	1937.131[a]	3	<.001
Likelihood Ratio	1468.279	3	<.001
Linear-by-Linear Association	1.687	1	.194
N of Valid Cases	9800		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 70.08.